

AKSHAT PANDEY

Backend Engineer — Python, Django, FastAPI · Video streaming, async systems & MLOps

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PROFESSIONAL EXPERIENCE

Backend Developer

June 2024 – Present

Intozi Tech. Pvt. Ltd. — Computer Vision & Video Analytics

Gurugram, India

- Own Python backend development across the company's products, including **Ikshana**, its core video-analytics platform, driving feature delivery and coordinating frontend work within a small team.
- Built the backend for a client-facing Video Management System: designed the **Django** services and integrated **MediaMTX** for **RTSP/WebRTC** streaming, surfacing live camera feeds and AI analytics in the app in real time.
- Architected the async layer on **PostgreSQL**, **Redis**, **RabbitMQ**, and **Celery**, moving video-processing and AI workloads off the request path to keep services responsive under load.
- Built an internal **MLOps pipeline** (Django + React) covering the full loop — dataset upload, auto-labeling, human verification, and model training/retraining — with a path to client-facing deployment.

PROJECTS

Papyrus 🗄 | *FastAPI, React, Celery, PostgreSQL, Redis, S3, Docker*

2026 – Present

- Self-hostable, privacy-first PDF toolkit — merge, split, compress, rotate, reorder, and OCR — served through a fast browser UI over an asynchronous **Celery** job pipeline.
- Async **FastAPI** backend (SQLAlchemy 2.x, asyncpg, Alembic) with **S3**-compatible storage and a zero-retention TTL purge; full stack containerized with Docker Compose and shipped with **Kubernetes/Helm** artifacts.

HeadTogether 🗄 | *FastAPI, React, WebSockets, Redis, SQLAlchemy*

2024 – 2025

- Social app built on **geo-bounded rooms** — ephemeral chat spaces discoverable only within a configurable radius, with distance-ranked discovery and coordinate rounding for non-member privacy.
- Backend spanning **73 REST routes** and 20 ORM tables (layered routes/services/repositories), real-time chat over **WebSockets** with Redis pub/sub fanout, and **JWT** auth with refresh-token rotation, Argon2id hashing, and rate limiting.

ShieldBuntu 🗄 | *FastAPI, React, Ansible, SQLite*

2024

- One-click Ubuntu hardening tool turning **CIS Benchmark** controls into **16 idempotent Ansible roles** (SSH, firewall, kernel sysctl, AppArmor, auditd, fail2ban, USB control), each with apply, dry-run, and snapshot revert.
- **FastAPI** daemon that discovers roles and exposes them as a typed HTTP API, streaming live Ansible events to a **React** UI via **Server-Sent Events**, with a concurrency-safe orchestrator and **PAM**-backed auth.

MeshHawk 🗄 | *FastAPI, React, scapy, NetworkX, ARQ, PostgreSQL, Redis*

2023 – 2024

- Local-first WiFi mesh detector that ingests **.pcap/.pcapng** captures, reconstructs 802.11 topology with **scapy** into a directed MAC graph in **NetworkX**, and classifies meshes via node-degree heuristics — no data leaves the machine.
- Async **FastAPI** backend with an **ARQ/Redis** worker pipeline, **PostgreSQL** (JSONB + GIN indexes), live monitor-mode capture, and on-demand PDF reports rendering each topology as inline SVG.

TECHNICAL SKILLS

Languages: Python, Bash, SQL

Frameworks: Django, FastAPI, Django Channels

Data & Messaging: PostgreSQL, Redis, RabbitMQ, Celery, SQLAlchemy

Infrastructure: Docker, Docker Compose, Kubernetes, Helm, Nginx, AWS, S3, Linux, Git

Backend: REST APIs, WebSockets, Server-Sent Events, JWT auth, microservices, async/ASGI, RTSP/WebRTC streaming

EDUCATION

Bhilai Institute of Technology

Durg, India

B.Tech, Computer Science — CPI: 9.68

Aug. 2020 – May 2024

ACHIEVEMENTS

KAVACH'23 Winner: First place in the inaugural nationwide KAVACH'23 cybersecurity hackathon organized by the Government of India.